हो. 👍 खाली पूर्ण **professional PDF-ready content** देतो. तुम्ही हे LibreOffice Writer मध्ये paste करून **Export as PDF** करू शकता.

मी तुमच्या आजच्या project work नुसार current status आणि पुढची दिशा समाविष्ट केली आहे. विशेषतः **ATM correction** नंतर अजून **rebuild** करायच्या **tables** स्पष्ट केल्या आहेत.

OPTION ANALYSIS

Technical Documentation & Working Synopsis

Project Path

/home/manish/trading_project/option_analysis/

Primary Focus

- NIFTY
- BANKNIFTY
- 1-minute Options Analysis
- FYERS-based historical data
- Futures-based IV calculation
- OI and option activity analysis
- Market-state and future trading-signal research

Current Development Stage

Research and Data Analysis System

Important:
This project is currently a research system. The existing analysis should not be treated as a validated trading strategy. More historical data and proper out-of-sample validation are required before any trading rule can be considered reliable.

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1. PROJECT OBJECTIVE

The objective of this project is to build a structured and modular options-analysis system for Indian index options.

The system is designed to combine:

Underlying / Futures Price

+
Option Price
+
Open Interest
+
OI Change
+
Option Activity
+
PCR
+
IV
+
IV Rank / Percentile
+
OI Concentration
+
Market Trend
↓
Market State
↓
Research / Trading Signal

The project is being developed in stages.

The current implementation focuses only on:

NIFTY
BANKNIFTY

Later, the architecture can be extended to:

FINNIFTY
MIDCPNIFTY
NIFTYNXT50
NIFTYFPI
Stock Options

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2. PROJECT RULES

The following rules have been established for this project.

Rule 1 - No Docker for now

The project is being developed directly on Ubuntu using Python virtual environment.

Rule 2 - Futures project is READ-ONLY

The existing:

~/trading_project/futures_project/

is an input source for analysis.

It should not be modified by the option-analysis project unless explicitly required.

Rule 3 - FYERS remains the primary source

The project will continue using the existing FYERS-based data architecture.

Rule 4 - No manual NSE bulk download

NSE historical contract-wise data was investigated because of the historical OI limitation.

However, the available contract-wise historical data is daily rather than the required 1-minute historical chain.

Therefore:

Do NOT switch the project to NSE.

Do NOT manually download thousands of contracts.

Rule 5 - Research first, trading later

The objective is not to immediately create a BUY/SELL system.

First:

Data
→ Analysis
→ Validation
→ Signal Research
→ Paper Trading
→ Only later Live Trading

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3. CURRENT PROJECT STRUCTURE

```
trading_project/
│
├── futures_project/
│
└── option_analysis/
```

```
│
├── database/
│
├── notebooks/
│
└── scripts/
```

The option-analysis project is intentionally separated from the existing futures project.

---

4. DATABASE ARCHITECTURE

4.1 index_options.db

Purpose:

Stores selected historical option data.

Current scope:

NIFTY
BANKNIFTY

Data includes:

Index
Option Symbol
CE / PE
Strike
Expiry
Datetime
Open
High
Low
Close
Volume
Open Interest

Current selection methodology:

Nearest expiry
+
ATM ±20 strikes
+
1-minute data

Current available period:

September 1, 2026
to
September 25, 2026

---

5. underlying_data.db

Contains underlying index data.

Currently:

NIFTY
BANKNIFTY

with 1-minute data.

It also contains:

INDIA VIX

The INDIA VIX historical data currently extends much further back than the NIFTY/BANKNIFTY underlying dataset.

Important distinction:

NIFTY/BANKNIFTY do NOT currently have five years of 1-minute underlying history in this project.

The current working option dataset is primarily the September 2026 period.

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6. fyers_style_iv.db

This database stores the FYERS-style implied volatility calculation.

The methodology uses:

Futures Price
+
Option Price
+
Strike
+
Expiry

+
Risk-free rate
↓
Black-76
↓
Implied Volatility

The project uses the futures formulation instead of the earlier spot-based experiment.

The spot-based IV experiment is not the final methodology.

---

7. BLACK-76 IV METHODOLOGY

The project uses Black-76 because the option is being valued using the corresponding futures price.

Conceptually:

Futures Price = F
Strike = K
Risk-free rate = r
Time to expiry = T
Volatility = σ

The model calculates the volatility that makes the theoretical option price equal to the observed option price.

The workflow is:

Observed Option Price
↓
Black-76
↓
Solve for IV
↓
Calculated IV
↓
Store in database

The project uses OTM options for the IV calculation.

For a given strike/expiry/timestamp:

CE IV
PE IV

are intended to represent the same underlying volatility estimate.

---

8. ATM IV

ATM is determined using the corresponding futures price.

Example concept:

Futures = 23,190

Available strikes:
23,100
23,150
23,200
23,250
23,300

The strike nearest to futures is selected:

ATM = 23,200

Similarly:

BANKNIFTY futures = 55,687.40
ATM = 55,700

This corrected futures-based ATM selection was an important change in the project.

---

9. IV RANK

IV Rank attempts to answer:

> Where is the current IV between its historical minimum and maximum?

Conceptually:

Historical Minimum IV
|
|
Current IV
|
|
Historical Maximum IV

A low IV Rank means current IV is closer to the historical low.

A high IV Rank means current IV is closer to the historical high.

However:

Current dataset contains only 18 trading days.

Therefore the current IV Rank is preliminary.

It should NOT be treated as a long-term volatility regime measure.

---

10. IV PERCENTILE

IV Percentile answers a slightly different question:

> What percentage of historical observations had IV at or below the current IV?

Example:

IV Percentile = 80

means approximately 80% of the available historical observations were at or below the current observation.

Again:

The current historical sample is only 18 trading days.

Therefore the percentile is currently a research feature rather than a robust long-term measure.

---

11. OPEN INTEREST ANALYSIS

The project calculates:

CE OI
PE OI
Total OI
CE OI Change
PE OI Change
Total OI Change

OI is analysed at:

Individual Strike
ATM
ATM ±1
ATM ±2
ATM ±5
ATM ±10
ATM ±20

---

12. OPTION ACTIVITY CLASSIFICATION

The project uses price change and OI change together.

Option Price	OI	Activity
UP	UP	LONG_BUILDUP
DOWN	UP	SHORT_BUILDUP
UP	DOWN	SHORT_COVERING
DOWN	DOWN	LONG_UNWINDING

Interpretation

LONG_BUILDUP

Price ↑
OI ↑

Interpretation:

New long positions are being added according to the classification framework.

SHORT_BUILDUP

Price ↓
OI ↑

Interpretation:

New short positions are being added according to the classification framework.

SHORT_COVERING

Price ↑
OI ↓

Interpretation:

Existing short positions are being closed.

LONG_UNWINDING

Price ↓
OI ↓

Interpretation:

Existing long positions are being closed.

Important

These are **market activity classifications**.

They are NOT automatically:

LONG = BUY
SHORT = SELL

They must be combined with other information and validated.

---

13. TOTAL PCR

The project calculates:

PCR = Total PE OI / Total CE OI

However, the current PCR is calculated only over the selected chain:

ATM ±20 strikes

Therefore it should be called:

Total PCR (Selected Chain)

and not simply "full-chain PCR".

Interpretation

PCR < 1

means selected CE OI is greater than selected PE OI.

PCR > 1

means selected PE OI is greater than selected CE OI.

PCR should not be used alone.

It should be interpreted with:

Price
+
OI Change
+
Activity
+
IV
+
Market Trend

---

14. OI CONCENTRATION

OI concentration measures how much OI exists around the current ATM.

The project examines:

ATM ±1
ATM ±2
ATM ±5
ATM ±10
ATM ±20

For example:

±5 concentration =

OI within ATM ±5

Selected chain total OI

This helps understand whether OI is concentrated near the current market price or spread farther away.

Important

High concentration near ATM does not automatically mean:

Support

or

Resistance

It is a contextual market-structure feature.

---

15. ACTIVITY BALANCE

The project calculates positive and negative activity.

Positive activity:

LONG_BUILDUP
+
SHORT_COVERING

Negative activity:

SHORT_BUILDUP
+
LONG_UNWINDING

Therefore:

Activity Balance
=
Positive Activity
-
Negative Activity

Positive balance means positive activity classifications dominate.

Negative balance means negative activity classifications dominate.

Again:

This is a research feature, not a validated trading signal.

---

16. MARKET TREND

The project examines futures movement over a short horizon.

Current research uses:

5-candle futures movement

For example:

Current Futures Price
-
Price 5 candles earlier

This creates:

Price Change
Price Change %

Market state can then be classified as:

UP
DOWN
FLAT

---

17. MARKET STATE

The project combines several dimensions:

Price State
+
CE OI State
+
PE OI State
+
PCR State
+
IV State
+
Activity Balance

Example conceptual state:

Price = UP
CE OI = INCREASE
PE OI = DECREASE

PCR = NORMAL
IV = MEDIUM

This creates a structured market-state observation.

The purpose is to study:

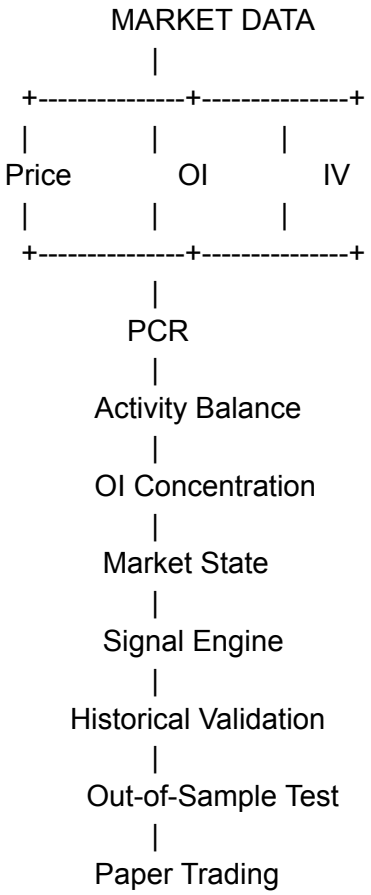
> What happened historically after this combination occurred?

---

18. TRADING SIGNAL - FUTURE DESIGN

The project should NOT immediately convert one indicator into a BUY/SELL signal.

Instead, the future signal engine should work like:



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19. EXAMPLE RESEARCH SIGNAL

A possible research condition could be:

Futures Price = UP

AND

CE Activity Balance = Positive

AND

PE Activity Balance = Negative

AND

PCR = Normal

AND

IV regime is stable

The system should NOT immediately say:

BUY

Instead it should record:

Signal State = Candidate Bullish State

Then measure:

5-candle return
10-candle return
15-candle return
30-candle return

and compare historical outcomes.

This allows us to determine whether the state actually contains useful information.

---

20. SIGNAL CONFIDENCE

A future signal engine can eventually combine multiple independent features.

For example:

Price Direction
+
OI Direction

+
Activity
+
PCR
+
IV
+
Concentration

Instead of:

ONE INDICATOR → TRADE

the system should eventually investigate:

MULTIPLE CONFIRMATIONS
↓
MARKET STATE
↓
HISTORICAL OUTCOME
↓
VALIDATED SIGNAL

But this validation has not yet been achieved.

---

21. CURRENT DATA LIMITATION

The biggest current limitation is the historical sample.

Current option data:

September 1–25, 2026

Trading days:

18

This is too small for establishing a reliable trading relationship.

The current results should therefore be considered:

Research observations

and not:

Proven strategy

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22. DEVELOPMENT / VALIDATION SPLIT

The project used a chronological split.

Development period

September 1–15, 2026

Validation period

September 16–25, 2026

The reason for chronological splitting is to avoid using future information while developing the research rules.

---

23. FORWARD 5-CANDLE ANALYSIS

The project measures future movement after the current market state.

For example:

Current candle
↓
Wait 5 candles
↓
Measure futures return

The calculation was corrected so that the forward return remains within the same trading day.

This prevents overnight/session-boundary contamination.

---

24. VALIDATION RESULT

The analysis produced recurring combinations of:

Price State
+
CE OI State
+
PE OI State
+

PCR State
+
IV State

Some combinations showed different historical average returns or UP/DOWN distributions.

However:

The current sample is only 18 trading days.

Therefore no reliable trading relationship should be concluded from these results.

The next requirement is:

More historical data
↓
More observations
↓
Development
↓
Out-of-sample validation
↓
Robustness testing

---

25. IMPORTANT DATA-CONSISTENCY ISSUE

This is currently one of the most important technical tasks.

The futures-based ATM selection was corrected.

The corrected ATM IV and ATM OI analysis were rebuilt.

Current latest corrected examples are:

BANKNIFTY Futures = 55,687.40
ATM = 55,700

NIFTY Futures = 23,190
ATM = 23,200

However, some downstream tables were originally generated before this correction.

Therefore these tables need to be rebuilt/validated:

atm_oi_concentration

atm_activity_summary
market_trend_analysis
combined_market_analysis
market_state_analysis

Some selected fields in `combined_market_analysis` were already updated from corrected ATM analysis, but all dependent fields have not yet been fully rebuilt.

Important

Do NOT use the current snapshot as the final production-quality analysis until this consistency issue is resolved.

---

26. CURRENT SNAPSHOT

The script:

scripts/current_option_snapshot.py

provides a reusable read-only market snapshot.

It currently reports:

Futures Price
5-candle Price Change
ATM Strike
CE OI
PE OI
Total PCR
CE ΔOI
PE ΔOI
Activity Balance
ATM IV
IV Rank
IV Percentile
OI Concentration

The script itself works.

But some concentration/activity/trend values may still be based on pre-correction downstream data.

Therefore:

Script = working
Data pipeline = needs final consistency rebuild

---

27. NEXT DEVELOPMENT ROADMAP

Phase 1 - Data consistency

First rebuild all ATM-dependent tables.

Correct Futures
↓
Correct ATM
↓
ATM OI
↓
OI Concentration
↓
Activity Summary
↓
Market Trend
↓
Combined Analysis
↓
Market State

Then run consistency checks.

Phase 2 - Historical data expansion

Increase the historical period substantially.

The current:

18 trading days

is not sufficient.

The objective should be to build a much larger historical research dataset before deciding whether any relationship exists.

Phase 3 - Feature validation

Test individual features:

Price
OI
ΔOI
PCR
IV
IV Rank
IV Percentile
Activity
OI Concentration

Determine which features contain useful information.

Phase 4 - Combined market states

Test combinations.

Example:

UP
+
CE OI INCREASE
+
PE OI DECREASE
+
PCR NORMAL
+
IV MEDIUM

Measure subsequent returns.

Phase 5 - Signal engine

Create a dedicated signal table.

Example:

datetime
index
market_state
signal
signal_strength
entry_price

forward_5c_return
forward_10c_return
forward_30c_return

The signal should initially be a research label.

---

28. PAPER TRADING - FUTURE PHASE

Only after historical validation:

Historical Research
↓
Out-of-Sample Validation
↓
Paper Trading
↓
Risk Testing
↓
Live Trading

Do not jump directly from historical pattern detection to live trading.

---

29. RISK MANAGEMENT - FUTURE

A future trading system must separately define:

Entry
Stop Loss
Target
Position Size
Maximum Daily Loss
Maximum Trades
Slippage
Brokerage
Taxes
Transaction Costs
Gap Risk
Liquidity

A signal can appear statistically useful but still fail after transaction costs or poor risk management.

Therefore signal quality and risk management should be tested separately.

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30. WHAT NOT TO DO

Do not:

Start another NSE bulk-download experiment

Do not:

Modify futures_project

Do not:

Treat current 18-day results as proof

Do not:

Use PCR alone as a trading signal

Do not:

Use OI activity classification alone as a BUY/SELL signal

Do not:

Build live trading before proper validation

Do not:

Mix corrected ATM data with old ATM-dependent analysis

---

31. PERSONAL DEVELOPER QUICK START

This section is for personal use and is intentionally not part of the GitHub source-code documentation.

After PC restart:

```
cd ~/trading_project
source .venv/bin/activate
which python
```

python --version
cd ~/trading_project/option_analysis
git status

Expected:

/home/manish/trading_project/.venv/bin/python

Python:

Python 3.13.11

---

32. IMPORTANT PROJECT FILES

Main scripts currently include:

scripts/build_index_option_master.py

scripts/download_index_options.py

scripts/download_underlying_data.py

scripts/calculate_fyers_style_iv.py

scripts/build_atm_iv.py

scripts/calculate_iv_rank_percentile.py

scripts/build_strike_oi_analysis.py

scripts/calculate_oi_change.py

scripts/calculate_oi_change_pct.py

scripts/calculate_option_price_change.py

scripts/classify_option_activity.py

scripts/build_atm_oi_analysis.py

scripts/current_option_snapshot.py

Testing scripts include:

scripts/test_daily_selection.py

scripts/test_fyers_style_iv.py

scripts/test_option_history.py

---

33. FINAL PROJECT STATUS

Completed

✓ Option contract master
✓ NIFTY option data
✓ BANKNIFTY option data
✓ Underlying data
✓ INDIA VIX
✓ Futures-based Black-76 IV
✓ ATM IV
✓ IV Rank
✓ IV Percentile
✓ Strike OI analysis
✓ OI change
✓ OI percentage
✓ Option price change
✓ Activity classification
✓ ATM OI analysis
✓ Selected-chain PCR
✓ OI concentration
✓ Activity balance
✓ Market trend
✓ Market state
✓ Forward 5-candle research
✓ Current option snapshot

Not yet final

⚠ ATM-dependent downstream tables need complete rebuild
⚠ Historical sample is too short
⚠ Trading signals are not validated
⚠ Paper trading has not started
⚠ Live trading has not started

---

34. CURRENT PRIORITY

The next task should be:

REBUILD AND VALIDATE ALL ATM-DEPENDENT TABLES

Specifically:

atm_oi_concentration
↓
atm_activity_summary
↓
market_trend_analysis
↓
combined_market_analysis
↓
market_state_analysis

After that:

Data consistency check
↓
Historical data expansion
↓
Validation
↓
Signal research

---

35. PROJECT PHILOSOPHY

The project should follow this principle:

DATA FIRST
↓
CLEAN DATA
↓
CORRECT CALCULATIONS
↓
MARKET FEATURES
↓
HISTORICAL RESEARCH
↓
OUT-OF-SAMPLE VALIDATION
↓
PAPER TRADING
↓
RISK MANAGEMENT
↓
LIVE TRADING

Never reverse this order.

The goal is not to find a signal quickly.

The goal is to determine, using clean historical data and proper validation, whether a market relationship actually exists.

END OF DOCUMENT

Document type: Personal Technical Working Synopsis
Project: Option Analysis
Primary instruments: NIFTY and BANKNIFTY
Current stage: Research / Data Analysis
Next technical priority: Rebuild and validate ATM-dependent downstream analysis tables.